

ICS1513 - Computer Networks Laboratory

Experiment 1: Study of networking diagnostic and monitoring tools and Socket programming system calls

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Question 1a: Apply and analyze the use of various networking diagnostic and monitoring tools in a computer network environment. Demonstrate the execution of the following tools and record your observations regarding network status, configuration, connectivity, routing, and performance:

- netstat
- ifconfig
- iwconfig
- ping
- route
- nmap
- tcpdump
- dig
- traceroute
- mtr
- curl
- wget
- tracepath
- nslookup
- host
- whois

```
cseal@wtl-13: ~  
=====br/>Executing: netstat -rn  
=====br/>Kernel IP routing table  
Destination      Gateway          Genmask          Flags   MSS Window  irtt Iface  
0.0.0.0          10.6.0.1         0.0.0.0          UG      0 0        0 eno1  
10.6.0.0         0.0.0.0          255.255.0.0      U       0 0        0 eno1  
169.254.0.0      0.0.0.0          255.255.0.0      U       0 0        0 eno1  
=====br/>Command finished. 5 seconds to screenshot...  
=====br/>]
```

Figure 1: netstat

```
cseal@wtl-13: ~  
=====br/>Executing: ifconfig  
=====br/>eno1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.6.13.13 netmask 255.255.0.0 broadcast 10.6.255.255  
    inet6 fe80::3755:7a95:dfe1:b87a prefixlen 64 scopeid 0x20<link>  
    ether 28:c5:c8:75:a3:92 txqueuelen 1000 (Ethernet)  
    RX packets 129754 bytes 102636457 (102.6 Mb)  
    RX errors 0 dropped 293 overruns 0 frame 0  
    TX packets 27753 bytes 8504418 (8.5 Mb)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
    device interrupt 19 memory 0x88900000-88920000  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 2751 bytes 302165 (302.1 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 2751 bytes 302165 (302.1 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
=====br/>Command finished. 5 seconds to screenshot...  
=====br/>]
```

Figure 2: ifconfig

```
cseal@wtl-13: ~  
=====br/>Executing: iwconfigbr/>=====br/>lo          no wireless extensions.  
eno1       no wireless extensions.  
  
=====br/>Command finished. 5 seconds to screenshot...  
=====br/>[]
```

Figure 3: iwconfig

```
cseal@wtl-13: ~  
=====br/>Executing: ping -c 4 google.com  
=====br/>PING google.com (192.178.193.113) 56(84) bytes of data:  
64 bytes from ly-in-f113.1e100.net (192.178.193.113): icmp_seq=1 ttl=113 time=23.3 ms  
64 bytes from ly-in-f113.1e100.net (192.178.193.113): icmp_seq=2 ttl=113 time=23.2 ms  
64 bytes from ly-in-f113.1e100.net (192.178.193.113): icmp_seq=3 ttl=113 time=23.2 ms  
64 bytes from ly-in-f113.1e100.net (192.178.193.113): icmp_seq=4 ttl=113 time=23.1 ms  
  
--- google.com ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3084ms  
rtt: min/avg/max/mdev = 23.141/23.192/23.252/0.039 ms  
  
=====br/>Command finished. 5 seconds to screenshot...  
=====br/>[]
```

Figure 4: ping

```
=====
Executing: route -n
=====

Kernel IP routing table
Destination    Gateway      Genmask      Flags Metric Ref    Use Iface
0.0.0.0        10.6.0.1     0.0.0.0      UG    100    0      0 eno1
10.6.0.0       0.0.0.0      255.255.0.0  U     100    0      0 eno1
169.254.0.0    0.0.0.0      255.255.0.0  U     1000   0      0 eno1

=====
Command finished. 5 seconds to screenshot...
=====
[]
```

Figure 5: route

```
=====
Executing: nmap -F localhost
=====

Starting Nmap 7.80 ( https://nmap.org ) at 2026-06-25 10:37 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000025s latency).
Not shown: 98 closed ports
PORT      STATE SERVICE
631/tcp   open ipp
3306/tcp   open mysql

Nmap done: 1 IP address (1 host up) scanned in 0.01 seconds

=====
Command finished. 5 seconds to screenshot...
=====
[]
```

Figure 6: nmap

```
cseal@wtl-13: ~  
=====br/>Executing: tcpdump -c 5 -i any  
=====br/>tcpdump: any: You don't have permission to capture on that device  
(socket: Operation not permitted)  
=====br/>Command finished. 5 seconds to screenshot...  
=====br/>[]
```

Figure 7: tcpdump

```
cseal@wtl-13: ~  
=====br/>Executing: dig google.com  
=====br/>; <<> DiG 9.18.28-Ubuntu0.22.04.1-Ubuntu <<> google.com  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 6536  
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1  
br/>;; OPT PSEUDOSECTION:  
;; EDNS: version: 0, flags: udp: 65494  
;; QUESTION SECTION:  
;google.com. IN A  
br/>;; ANSWER SECTION:  
google.com. 222 IN A 192.178.193.102  
google.com. 222 IN A 192.178.193.139  
google.com. 222 IN A 192.178.193.138  
google.com. 222 IN A 192.178.193.101  
google.com. 222 IN A 192.178.193.100  
google.com. 222 IN A 192.178.193.113  
br/>;; Query time: 0 msec  
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)  
;; WHEN: Thu Jun 25 10:37:23 IST 2026  
;; MSG SIZE rcvd: 135  
br/>=====br/>Command finished. 5 seconds to screenshot...  
=====br/>[]
```

Figure 8: dig

```
cseal@wtl-13: ~  
=====br/>Executing: traceroute -n 10 google.com  
=====br/>traceroute to google.com (192.178.193.113), 10 hops max, 60 byte packets  
1  _gateway (10.0.0.1)  0.427 ms  0.422 ms  0.427 ms  
2  10.100.0.2 (10.100.0.2)  0.415 ms  0.400 ms  0.396 ms  
3  10.25.1.10 (10.25.1.10)  0.394 ms  0.391 ms  0.388 ms  
4  182.74.172.121 (182.74.172.121)  11.526 ms  6.297 ms  11.520 ms  
5  182.79.243.201 (182.79.243.201)  5.662 ms  182.79.236.125 (182.79.236.125)  5.679 ms  72.14.197.130 (72.14.197.130)  1.788 ms  
6  * * *  
7  * * *  
8  142.251.252.95 (142.251.252.95)  18.592 ms  142.251.55.90 (142.251.55.90)  6.729 ms  172.253.64.227 (172.253.64.227)  22.291 ms  
9  142.251.51.118 (142.251.51.118)  5.708 ms  5.937 ms  142.251.50.58 (142.251.50.58)  6.261 ms  
10  192.178.82.107 (192.178.82.107)  23.922 ms  142.251.241.167 (142.251.241.167)  30.544 ms  142.251.252.95 (142.251.252.95)  31.148 ms  
=====br/>Command finished. 5 seconds to screenshot...  
=====br/>[
```

Figure 9: traceroute

```
cseal@wtl-13: ~  
=====br/>Executing: mtr -c 4 -r google.com  
=====br/>Start: 2020-06-25T10:37:46+0530  
HOST: wtl-13  
=====br/>1. |-- _gateway 0.0% 4 0.7 0.8 0.5 1.5 0.4  
2. |-- 10.100.0.2 0.0% 4 0.5 0.5 0.4 0.7 0.1  
3. |-- 10.25.1.10 0.0% 4 0.3 0.3 0.3 0.3 0.0  
4. |-- 182.74.172.121 0.0% 4 5.2 6.1 5.1 9.1 2.0  
5. |-- 110.119.101.149 0.0% 4 5.5 5.0 5.5 5.8 0.1  
6. |-- 74.125.147.238 0.0% 4 7.3 7.2 7.1 7.4 0.1  
7. |-- 142.251.71.187 0.0% 4 6.7 6.2 5.9 6.7 0.3  
8. |-- 172.253.71.2 0.0% 4 6.7 7.0 6.6 7.7 0.5  
9. |-- ??? 100.0 4 0.0 0.0 0.0 0.0 0.0  
10. |-- 192.178.254.30 0.0% 4 36.0 36.4 36.0 37.0 0.5  
11. |-- 192.178.82.105 0.0% 4 33.6 33.7 33.6 33.8 0.1  
12. |-- 142.251.254.182 0.0% 4 33.6 33.6 33.5 33.8 0.1  
13. |-- ??? 100.0 4 0.0 0.0 0.0 0.0 0.0  
14. |-- ??? 100.0 4 0.0 0.0 0.0 0.0 0.0  
15. |-- ??? 100.0 4 0.0 0.0 0.0 0.0 0.0  
16. |-- ??? 100.0 4 0.0 0.0 0.0 0.0 0.0  
17. |-- ??? 100.0 4 0.0 0.0 0.0 0.0 0.0  
18. |-- ??? 100.0 4 0.0 0.0 0.0 0.0 0.0  
19. |-- ly-in-f113.1e100.net 0.0% 4 30.6 31.3 30.5 33.6 1.5  
=====br/>Command finished. 5 seconds to screenshot...  
=====br/>[
```

Figure 10: mtr

```
cseal@wtl-13: ~  
=====br/>Executing: curl -I https://www.google.com  
=====br/>HTTP/2 429  
content-length: 1183  
content-type: text/html; charset=UTF-8  
date: Thu, 25 Jun 2026 05:08:07 GMT  
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000  
  
=====br/>Command finished. 5 seconds to screenshot...  
=====br/>[
```

Figure 11: curl

```
cseal@wtl-13: ~  
=====br/>Executing: wget --spider https://www.google.com  
=====br/>Spider mode enabled. Check if remote file exists.  
--2026-06-25 10:38:12-- https://www.google.com/  
Resolving www.google.com (www.google.com)... 142.251.151.119, 142.251.152.119, 142.251.157.119, ...  
Connecting to www.google.com (www.google.com)|142.251.151.119|:443... connected.  
HTTP request sent, awaiting response... 204 No Content  
Remote file exists.  
  
=====br/>Command finished. 5 seconds to screenshot...  
=====br/>[
```

Figure 12: wget

```
=====
Executing: tracepath -m 10 google.com
=====
1?: [LOCALHOST] pmtu 1500
1:  _gateway 0.703ms asymm 2
1:  _gateway 0.502ms asymm 2
2:  10.106.0.2 0.475ms asymm 3
3:  10.25.1.10 0.357ms
4:  122.15.35.174 1.840ms
5:  110.119.101.183 5.602ms asymm 6
6:  no reply
7:  no reply
8:  no reply
9:  no reply
10: no reply
Too many hops: pmtu 1500
Resume: pmtu 1500
=====
Command finished. 5 seconds to screenshot...
=====
```

Figure 13: tracepath

```
=====
Executing: nslookup google.com
=====
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.193.101
Name:   google.com
Address: 192.178.193.113
Name:   google.com
Address: 192.178.193.100
Name:   google.com
Address: 192.178.193.139
Name:   google.com
Address: 192.178.193.102
Name:   google.com
Address: 192.178.193.138
Name:   google.com
Address: 2404:6800:4000:1017::65
Name:   google.com
Address: 2404:6800:4000:1017::66
Name:   google.com
Address: 2404:6800:4000:1017::8a
Name:   google.com
Address: 2404:6800:4000:1017::64
=====
Command finished. 5 seconds to screenshot...
=====
```

Figure 14: nslookup


```
cseal@wtl-13: ~  
===== Executing: host google.com =====  
=====  
google.com has address 192.178.193.100  
google.com has address 192.178.193.102  
google.com has address 192.178.193.139  
google.com has address 192.178.193.138  
google.com has address 192.178.193.113  
google.com has address 192.178.193.101  
google.com has IPv6 address 2404:6800:4000:1017::65  
google.com has IPv6 address 2404:6800:4000:1017::8a  
google.com has IPv6 address 2404:6800:4000:1017::64  
google.com has IPv6 address 2404:6800:4000:1017::66  
google.com mail is handled by 10 smtp.google.com.  
===== Command finished. 5 seconds to screenshot... =====  
=====
```

Figure 15: host

```
cseal@wtl-13: ~  
===== Executing: whois google.com =====  
=====  
Domain Name: GOOGLE.COM  
Registry Domain ID: 2138514_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.markmonitor.com  
Registrar URL: http://www.markmonitor.com  
Updated Date: 2019-09-09T15:39:04Z  
Creation Date: 1997-09-15T04:00:00Z  
Registry Expiry Date: 2028-09-14T04:00:00Z  
Registrar: MarkMonitor Inc.  
Registrar IANA ID: 292  
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com  
Registrar Abuse Contact Phone: +1.2086851750  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited  
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited  
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited  
Name Server: NS1.GOOGLE.COM  
Name Server: NS2.GOOGLE.COM  
Name Server: NS3.GOOGLE.COM  
Name Server: NS4.GOOGLE.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2020-06-25T05:08:32Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration.  
  
TERMS OF USE: You are not authorized to access or query our Whois  
database through the use of electronic processes that are high-volume and  
automated except as reasonably necessary to register domain names or  
modify existing registrations; the Data in VeriSign Global Registry  
Services' ("VeriSign") Whois database is provided by VeriSign for  
information purposes only, and to assist persons in obtaining information  
about or related to a domain name registration record. VeriSign does not  
guarantee its accuracy. By submitting a Whois query, you agree to abide  
by the following terms of use: You agree that you may use this Data only  
for lawful purposes and that under no circumstances will you use this Data  
to: (1) allow, enable, or otherwise support the transmission of mass  
unsolicited, commercial advertising or solicitations via e-mail, telephone,  
or facsimile; or (2) enable high volume, automated, electronic processes  
that apply to VeriSign (or its computer systems). The compilation,  
repackaging, dissemination or other use of this Data is expressly  
prohibited without the prior written consent of VeriSign. You agree not to  
use electronic processes that are automated and high-volume to access or  
query the Whois database except as reasonably necessary to register
```

Figure 16: whois